02-01

再平衡时间 $\tau \sim 10^{-15} \sim 10^{-12}$ s

02-01

例 2.1.7

(1) $q_A' = -q_B$　　$q_B' = q_B$

(2) $q_B'' = q_A + q_B$

02-04

问题：为什么 $W_{surface}$ 是导体单位体积的静电能

$$W_s = \frac{1}{2}\varepsilon_0\left(\frac{\sigma}{\varepsilon_0}\right)^2 = \frac{1}{2}\varepsilon_0 E_s^2 = \frac{\sigma^2}{2\varepsilon_0}$$

02-06

P65 例1

$$\begin{cases} \sigma_1 = \sigma_2 + \sigma_3 + \sigma_4 \\ \sigma_4 = \sigma_1 + \sigma_2 + \sigma_3 \\ \sigma_1 + \sigma_2 = \frac{Q}{S} \\ \sigma_3 + \sigma_4 = \frac{Q}{S} \end{cases}$$

解得 $\sigma_1 = \sigma_4 = \frac{1}{2}(Q+Q')/S$

$\sigma_2 = -\sigma_3 = \frac{1}{2}(Q-Q')/S$

02-06

例 2.1.1

(1) $F_{21} = \frac{1}{4\pi\varepsilon_0}\frac{q_1q_2}{r^2}$

(2) $F_2 = 0$

例 2.1.3

(1) $F = \frac{1}{4\pi\varepsilon_0}\frac{q_1q_2}{r^2}$　　$a = 0$

(2) $F' = F$　　$a' \neq 0$

例 2.1.10

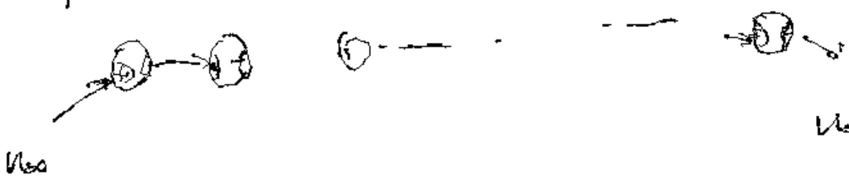
$\nabla^2 u = -\nabla\cdot E = \frac{\rho}{\varepsilon_0}$

极大/极小 → $\nabla^2 u > 0$ 或 < 0

略 - P25

证：若全部球均带负电荷.

则



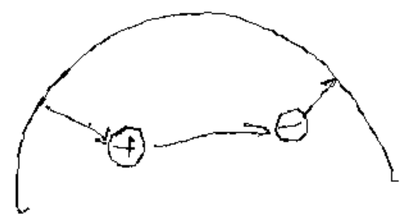
$u_\infty > u_1 >$ —— $> u_{1998} > u_0'$ 矛盾.

例 2.1.12

讨论于球壳

⇒ $u_+ > u_-$ （由Gauss定理）

$u_- > u_+$ （Gauss ~，和把2q，−q包含在内之Gauss面）



例 2.1.13

同 2.1.12

最后取大大大大大 Gauss面即可.

例 2.1.14

$u = u_q + u_{q'} = \frac{1}{4\pi\varepsilon_0}\frac{q}{a} + \int\frac{dq'}{4\pi\varepsilon_0 R} = \frac{q}{4\pi\varepsilon_0 a}$

$\int dq' = 0$

例 2.1.15

$u = \sum u_{q_i} + u_{q'_i} = \sum u_{q_i}$

例 2.1.16

$\frac{Q}{4\pi\varepsilon_0 R} + \frac{q}{4\pi\varepsilon_0 a} = 0 \Rightarrow Q = -\frac{R}{a}q$

例 2.1.20

(1) 设为 σ_0

$E_0 = \frac{\sigma_0}{\varepsilon_0}$

$E_0 \cdot 2\pi r_0 l = E(r) \cdot 2\pi r l \quad \to \quad E(r) = \frac{r_0}{r} E_0$

$\Delta U = \int_{r_0}^{r_1} E(r)\,dr = E_0 r_0 \ln\left(\frac{r_1}{r_0}\right)$

$\to E_0 =$ $\qquad\qquad$ $\sigma_0 =$

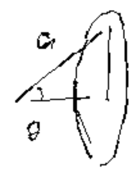
(2) $E_0 = c E_0$

例 2.1.23

$\theta_{b1} = \arccos\left(\frac{c}{a}\right) \qquad \theta_{b2} = \arccos\left(\frac{c}{b}\right)$

$\sigma_1 = \frac{-q}{4\pi a^2} \qquad \sigma_2 = \frac{q}{4\pi b^2}$

由Gauss定理可以证明. $E_1 = -\frac{\sigma_1}{2\varepsilon_0}$, $E_2 = \frac{\sigma_2}{2\varepsilon_0}$



$F_1 = -\int_0^{\theta_{b1}} \sigma_1 \cdot 2\pi a \sin\theta \cdot a\,d\theta \cdot E_1 \cdot \cos\theta$

$= \frac{-\pi a^2 q^2}{16\pi^2 \varepsilon_0 a^4} \int_0^{\theta_{b1}} \sin\theta\cos\theta\,d\theta$

$= \frac{-q^2}{16\pi\varepsilon_0 a^2} \cdot -\frac{1}{4}\cos(2\theta)\Big|_0^{\theta_{b1}}$

$= \frac{-q^2}{64\pi\varepsilon_0 a^2}\left[\left(2\cdot\frac{c^2}{a^2}-1\right)-1\right] = -\frac{q^2}{32\pi\varepsilon_0 a^2}\frac{c^2-a^2}{a^2}$

$F_2 = \int_0^{\theta_{b2}} \sigma_2 \cdot 2\pi b\sin\theta \cdot b\,d\theta \cdot E_2 \cdot \cos\theta$

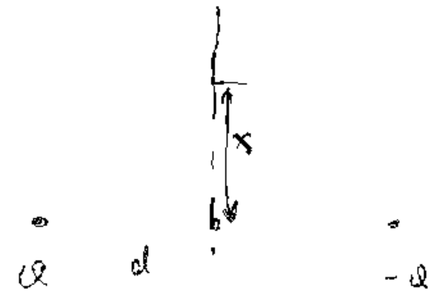
$= \frac{q^2}{32\pi\varepsilon_0 b^2}\frac{c^2-b^2}{b^2}$

$F_1 + F_2 = \frac{q^2}{32\pi\varepsilon_0}\left(\frac{c^2-b^2}{b^4} - \frac{c^2-a^2}{a^4}\right) = \frac{q^2}{32\pi\varepsilon_0 a^4 b^4}\left(c^2a^4 - b^2a^4 - c^2b^4 + a^2b^4\right)$

$= \frac{q}{32\pi\varepsilon_0 a^4 b^4}(a^2-b^2)\left[c^2(a^2+b^2) - a^2b^2\right] < 0 \quad \Rightarrow c < \frac{ab}{[illegible]}$

02-06

P68 例1

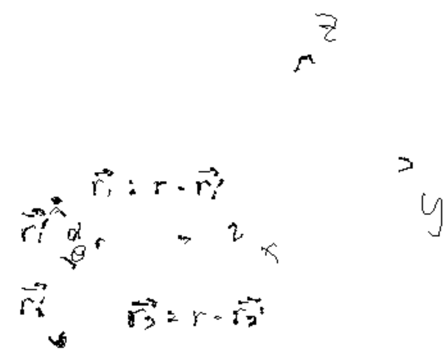


$$E(x) = -\frac{Q}{4\pi\varepsilon_0}\cdot\frac{2d}{(x^2+d^2)^{3/2}}$$

$$= \frac{\sigma(x)}{2\varepsilon_0}$$

$$\Rightarrow \sigma(x) = -\frac{Q}{\pi}\frac{d}{(x^2+d^2)^{3/2}}$$

02-07 镜像问题



1. 求一对电偶极子在空间中的 E, V 分布。

$$V(r) = \frac{q}{4\pi\varepsilon_0}\left(\frac{1}{r_1}-\frac{1}{r_2}\right)$$

$$= \frac{q}{4\pi\varepsilon_0}\left\{\frac{1}{[x^2+y^2+(z-d)^2]^{1/2}} - \frac{1}{[x^2+y^2+(z+d)^2]^{1/2}}\right\}$$

$$= \frac{q}{4\pi\varepsilon_0}\left[\frac{1}{(r^2+d^2-2rd\cos\theta)^{1/2}} - \frac{1}{(r^2+d^2+2rd\cos\theta)^{1/2}}\right]$$

$$\vec{E} = -\nabla\cdot V(r) = -\left(\frac{\partial V}{\partial r}\hat{r} + \frac{1}{r}\frac{\partial V}{\partial\theta}\hat{\theta} + \frac{1}{r\sin\theta}\frac{\partial V}{\partial\varphi}\hat{\varphi}\right)$$

$$\frac{\partial V}{\partial r} = \frac{q}{4\pi\varepsilon_0}\left(-\frac{1}{2}\right)\cdot\left[\frac{2r-2d\cos\theta}{(r^2+d^2-2rd\cos\theta)^{3/2}} - \frac{2r+2d\cos\theta}{(r^2+d^2+2rd\cos\theta)^{3/2}}\right]$$

$$= -\frac{q}{4\pi\varepsilon_0}\left[\frac{r-d\cos\theta}{(r^2+d^2-2rd\cos\theta)^{3/2}} - \frac{r+d\cos\theta}{(r^2+d^2+2rd\cos\theta)^{3/2}}\right]$$

$$\frac{1}{r}\frac{\partial V}{\partial\theta} = \frac{q}{4\pi\varepsilon_0 r}\left(-\frac{1}{2}\right)\left[\frac{2rd\sin\theta}{(r^2+d^2-2rd\cos\theta)^{3/2}} - \frac{-2rd\sin\theta}{(r^2+d^2+2rd\cos\theta)^{3/2}}\right]$$

$$= -\frac{qd\sin\theta}{4\pi\varepsilon_0}\left[\frac{1}{(r^2+d^2-2rd\cos\theta)^{3/2}} + \frac{1}{(r^2+d^2+2rd\cos\theta)^{3/2}}\right]$$

$$\frac{1}{r\sin\theta}\frac{\partial V}{\partial\varphi} = 0$$

2. 求镜面处导体板电荷密度分布。

$$n(r, \tfrac{\pi}{2}, 0) = \hat{z}$$

$$E_n(r, \tfrac{\pi}{2}, 0) = \frac{1}{r}\frac{\partial V}{\partial\theta} = -\frac{qd\sin\theta}{4\pi\varepsilon_0}\left[\frac{1}{(r^2+d^2-2rd\cos\theta)^{3/2}} - \frac{1}{(r^2+d^2+2rd\cos\theta)^{3/2}}\right]$$

$$= -\frac{qd}{\ldots}\cdot\frac{1}{(r^2+d^2)^{3/2}}$$

$$2\pi\varepsilon_0 \qquad r+d$$

$$\sigma(r, \tfrac{\pi}{2}, 0) = \varepsilon_0 \cdot E_n = -\frac{q}{2\pi} \cdot \frac{d}{(r^2+d^2)^{3/2}}$$

验证：

$$\int_0^\infty \sigma \cdot 2\pi r \cdot dr = -q \int_0^\infty \frac{d\, r\, dr}{(r^2+d^2)^{3/2}}$$

$$= -qd \cdot \left. -(r^2+d^2)^{-\frac{1}{2}} \right|_{r=0}^{r=\infty} = -q\, d \cdot \left(\frac{1}{d} - 0\right) = -q$$

3. 求导体板与 q 之间的作用

$$dE_q(r) = \frac{\sigma \cdot 2\pi r\, dr}{4\pi\varepsilon_0} \cdot \frac{d}{(r^2+d^2)^{3/2}} = \frac{\sigma r}{2\varepsilon_0} \frac{d}{(r^2+d^2)^{3/2}}\, dr$$

$$= -\frac{q}{4\pi\varepsilon_0} \frac{d^2 r}{(r^2+d^2)^3} \cdot dr$$

$$E_q = \int_0^\infty dE_q(r) = -\frac{q d^2}{4\pi\varepsilon_0} \int_0^\infty \frac{r\, dr}{(r^2+d^2)^3}$$

$$= -\frac{q d^2}{4\pi\varepsilon_0} \left. \frac{1}{4} - (r^2+d^2)^{-2} \right|_{r=0}^{r=\infty} = -\frac{q}{4\pi\varepsilon_0 (2d)^2}$$

$$F = q E_q$$

$$= -\frac{q^2}{4\pi\varepsilon_0 (2d)^2}$$ 即为电像法所得结果

4. 功 / 能量.

$$F(z) = -\frac{q^2}{4\pi\varepsilon_0 (2z)^2}$$

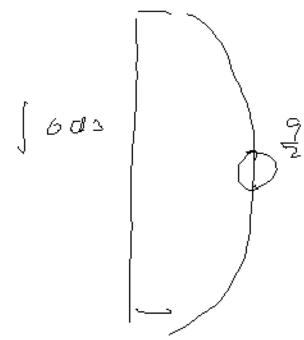
$$W = \int_\infty^d -F(z)\, d(-z) = \frac{q^2}{16\pi\varepsilon_0} \int_\infty^d \frac{dz}{z^2} = -\frac{q^2}{16\pi\varepsilon_0 d}$$

02-07

例 2.1.44/45

(1) 见上

(2) $\int_0^l \sigma(r)\, dS$



$$= \int_0^l -\frac{q}{2\pi} \cdot \frac{d}{(r^2+d^2)^{3/2}}\, 2\pi r\, dr$$

$$= -qd\int_0^l \frac{r\,dr}{(r^2+d^2)^{3/2}} = -qd \cdot -(r^2+d^2)^{-\frac{1}{2}}\Big|_{r=0}^{r=l}$$

$$= qd\left[\frac{1}{(l^2+d^2)^{1/2}} - \frac{1}{d}\right] = -\frac{1}{2}q$$

解得 $l = \sqrt{3}d$

亦可以暴力解决　本题：例 2.1.45

$$u(r) = \frac{q}{4\pi\varepsilon_0}\left(\frac{1}{r_1} - \frac{1}{r_2}\right)$$

$$= \frac{q}{4\pi\varepsilon_0}\left\{\frac{1}{[x^2+y^2+(z-d)^2]^{1/2}} - \frac{1}{[x^2+y^2+(z+d)^2]^{1/2}}\right\}$$

$$\vec{E} = -\nabla u = -\frac{q}{4\pi\varepsilon_0}\left(\left\{\frac{-x}{[x^2+y^2+(z-d)^2]^{3/2}} - \frac{-x}{[x^2+y^2+(z+d)^2]^{3/2}}\right\},\right.$$

$$\left.\left\{\frac{-y}{[x^2+y^2+(z-d)^2]^{3/2}} - \frac{-y}{[x^2+y^2+(z+d)^2]^{3/2}}\right\}, \left\{\frac{-(z-d)}{[x^2+y^2+(z-d)^2]^{3/2}} - \frac{-(z+d)}{[x^2+y^2+(z+d)^2]^{3/2}}\right\}\right)$$

$x = 0$

$$= \left(y\left\{\frac{1}{[y^2+(z-d)^2]^{3/2}} - \frac{1}{[y^2+(z+d)^2]^{3/2}}\right\}, \frac{z-d}{[y^2+(z-d)^2]^{3/2}} - \frac{z+d}{[y^2+(z+d)^2]^{3/2}}\right)$$

$$\frac{dz}{dy} = \frac{E_z}{E_y}$$

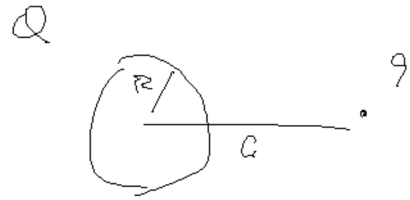
$$= \frac{(z-d)[y^2+(z+d)^2]^{3/2} - (z+d)[y^2+(z-d)^2]^{3/2}}{y\{[y^2+(z+d)^2]^{3/2} - [y^2+(z-d)^2]^{3/2}\}} = 0$$

$$d \quad dz = \frac{-d(l^2+d^2)^{3/2} - d(l^2+d^2)^{3/2}}{\ldots}$$

$$\lim_{\substack{y\to 1\\ z\to 0}} \quad \frac{\quad}{\quad} \quad \frac{\quad}{\{ (l^2+a^2)^{3/2} - (l^2+a^2)^{3/2} \}}$$

02-07

附加问题 2.1.46

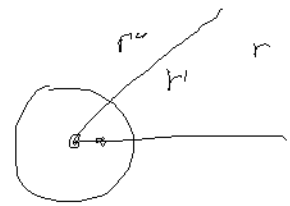


$$q' = -\frac{R}{a} q \qquad d = \frac{R^2}{a}$$

$$q'' = Q - q'$$

$$U_{球面} = \frac{q''}{4\pi\varepsilon_0 R} + \frac{q'}{4\pi\varepsilon_0 r'} + \frac{q}{4\pi\varepsilon_0 r} = \frac{Q + \frac{R}{a}q}{4\pi\varepsilon_0 R}$$

球外一点电势



$$U = \frac{1}{4\pi\varepsilon_0}\left(\frac{q}{r} + \frac{Q + q\frac{R}{a}}{r''} + \frac{q'}{r'}\right)$$

点电荷 q 所受力

$$F = \frac{1}{4\pi\varepsilon_0}\left[\frac{qq''}{a^2} + \frac{qq'}{(a-\frac{R^2}{a})^2}\right]$$

$$= \frac{q}{4\pi\varepsilon_0}\left[\frac{Q + q\frac{R}{a}}{a^2} - \frac{q\,aR}{(a^2-R^2)^2}\right]$$

$q \gg Q$ 时 / $a \to R$ 时

$F < 0$

唯一性定理：

给定：电荷分布（每个导体上的总电荷量 Q_k） 唯一确定 电势分布

电势分布（每个导体的电势 U_k） 电荷分布

若空间中无电荷 $\nabla^2 U=0$，则电势为一常数

假设 $\exists\ U_1(\vec{r}),\ U_2(\vec{r})$ 两分布

$\hookrightarrow\ \nabla^2 U_1=0 \quad \nabla^2 U_2=0 \qquad U_1=U_2$（在 ∂V 上）

令 $V_d=U_1-U_2$ 则 $\nabla^2 V_d=0$，$V_d=0 \quad (\partial V)$

令 $\vec{A}=V_d\nabla V_d$

$$\int_V \nabla\cdot\vec{A}\,dV=\oint_{\partial V}\vec{A}\cdot d\vec{s}$$

$$LHS=\int_V\left[(\nabla V_d)^2+V_d\nabla^2 V_d\right]dV$$

$$=\int_V(\nabla V_d)^2\,dV$$

$$=RHS=\oint_{\partial V}V_d\nabla V_d\cdot d\vec{S}=0$$

$\Rightarrow\ \nabla V_d=0\ \rightarrow\ V_d=\text{Const.}$

U_1, U_2 只相差一个常数，本质上是相同的电势分布.

两类边界条件：

(1) Dirichlet

每个导体上的电势固定 → 解唯一

(2) Neumann

每个导体表面电场固定 → ∇U 固定 → 解差常数

02-09

예 p99 -2.7)

$$E = \frac{\sigma}{\varepsilon_0} = \frac{u}{d}$$

$$d = d_0\left(1+\frac{\theta x}{d_0}\right) \quad \Rightarrow \quad \sigma = \frac{\varepsilon_0}{d_0}\,\frac{u}{1+\frac{\theta x}{d_0}}$$

$$\int \sigma a\,dx = \frac{\varepsilon_0 u a}{d_0}\int_0^a \frac{dx}{1+\frac{\theta x}{d_0}} = \frac{\varepsilon_0 u a}{\theta}\ln\left(1+\frac{\theta a}{d_0}\right) = Q$$

$$\Rightarrow u = \frac{Q\theta}{a\varepsilon_0}\cdot\frac{1}{\ln\left(1+\frac{\theta a}{d_0}\right)}$$

$$\sigma(x) = \frac{Q\theta}{a}\cdot\frac{1}{(d_0+\theta x)\ln\left(1+\frac{\theta x}{d_0}\right)}$$

$$C = \frac{Q}{u} = \frac{a\varepsilon_0}{\theta}\ln\left(1+\frac{\theta a}{d_0}\right)$$

02-10

$W = \frac{1}{2}\sum Qu$

$= \frac{1}{2}\int \rho u \, dV + \frac{1}{2}\int \sigma u \, dS$

静电能→聚集晶体中正负离子 → 晶体稳定

包剂1不相容

$w = \frac{1}{2}\varepsilon_0 E^2$

$W = \frac{1}{2}\int_V \varepsilon_0 E^2 \, dV$

电荷极2相互作用能

$W = -\vec{p_2}\cdot\vec{E_1} = \vec{p}\cdot\frac{1}{4\pi\varepsilon_0 r^5}[3(\vec{p_1}\cdot\vec{r})\vec{r} - r^2\vec{p_1}]$

$= \frac{1}{4\pi\varepsilon_0 r^5}[3(\vec{p_1}\cdot\vec{r})(\vec{p_2}\cdot\vec{r}) - r^2\vec{p_1}\cdot\vec{p_2}]$

三种电势能计算方法及物理意义

① $\int u \, dq$

外力将 dq 从 $u=0$ 处移至 $u=u$ 处所做的功

= 静电能增量.

② $\frac{1}{2}\sum Qu$

电荷分布之边静态

③ w

问：为什么 $\int_V dw = \frac{1}{2}\varepsilon_0 E^2 \, dV \geq 0$

而 $\frac{1}{2}\sum Qu$ 可能 >0，可能 <0

↳ $\int_V dW = \frac{1}{2}\sum QU$ → 自能

02-10

例 3.22.

$$\underbrace{W}_{\int \frac{1}{2}\varepsilon_0 E^2 dV} = \underbrace{W_{内自能} + W_{外自能}}_{QU_s} + \underbrace{W_{相互作用能}}_{\frac{1}{2}QU}$$

U_s 为自身电荷在自身处产生的电势大小.

$\frac{3Q^2}{20\pi\varepsilon_0 R}$ 对球，均匀分布.

$\frac{Q^2}{8\pi\varepsilon_0 R}$ 对球、分布于表面.

02-14 库仑定律验证：